

Assignment 5

Information Retrieval and Text Mining 25/26*
irtm-teachers@ims.uni-stuttgart.de

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Discussion Session: 2026-02-05

- **Groups:** Working in groups of up to three people is encouraged, up to four people is allowed. More people are not allowed. Copying results from one group to another (or from elsewhere) is not allowed. Changing groups during the term is allowed.
- **Grading:** Passing the assignments is a requirement for participation in the exam in all modules IRTM can be part of. Altogether 80 points need to be reached. There are five assignments with 20 pen & paper points and 10 programming points each. That means, altogether, 150 points can be reached.
- **Submission:** Create a group in Ilias and submit a single ZIP file, which should contain:
 - A PDF file with all written (non-programming) parts of the assignment.
 - One or more .py files containing your code for the programming tasks.

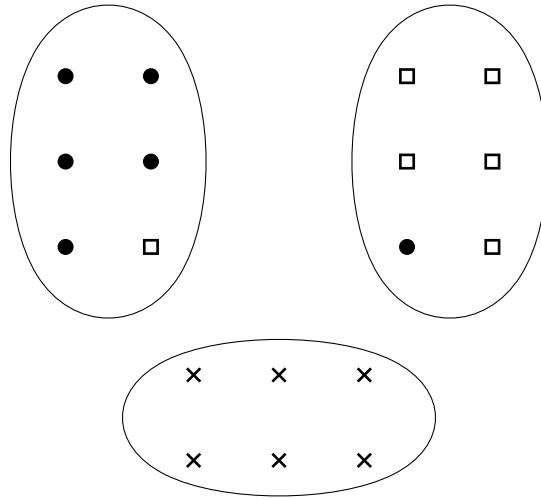
Include all group members' names on the first PDF page and as a comment in each Python file. If you are technically unable to create a group on Ilias (which occasionally happens), only one member should submit the ZIP file.

- **Make it understandable:** Do the best you can such that we can understand what you mean. Explain your solutions, comment and structure your code. Write your solutions in a way we can read them.
- **Handwriting:** We typically receive some submissions which are handwritten. That is fine, but if you submit handwritten solutions, make sure that they are well organized, easy to read and to understand, and that there is not doubt about the interpretation of letters. If you think that this might be hard, please typeset the solutions with a computer. We might reduce points if it's really tough for us and cannot read your submission properly.
- **Language:** Please submit your solutions in English.

*based on Information Retrieval and Text Mining 2023/24 by Roman Klinger

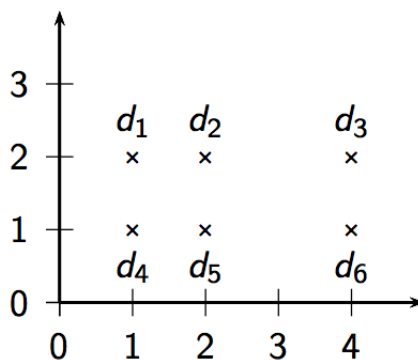
Task 1 (Evaluation of Clustering) 6 points

What is the purity of the following clustering, assuming that cross, circle and box are gold annotations of classes?



Task 2 (k Means) 4 points

Given the following instances (same example as in the lecture)

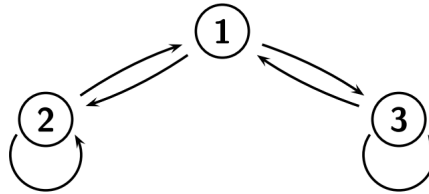


What is the RSS for the clustering that you achieve with the seeds d_2 and d_5 after convergence of the k means clustering? What is the RSS when you use the seeds d_2 and d_3 ? Explain your solution.

Task 3 (PageRank) 10 points

Subtask 3.1 Transition probability matrix (5 points)

Given the graph:



Calculate the transition probability matrix with teleportation rate $\alpha = 0.15$.

Subtask 3.2 PageRank calculation (5 points)

Calculate the PageRank values for the graph and transition matrix from Subtask 3.1. Explain all calculation steps and clearly highlight your final result. What is the name of the method that you used? Hint: if you start from a uniform distribution, calculating one step might be sufficient.

Programming Task (Clustering) 10 points

Implement k means clustering (from scratch, without using a library that implements clustering) and apply it to one of the document collections provided in this class (or a subset). Implement the calculation of RSS per cluster.

- Apply your implementation with $k = 2$ and with $k = 20$. For each clustering, what are the RSS values for each cluster? When you manually introspect the clusters with very low RSS, does that show in the documents that have been put into this cluster? Can you show some examples to exemplify your observations?
- Provide the list of documents which is closest to the cluster centers for $k = 20$ and discuss why these documents did not end up in the same clusters? Can you observe differences between the documents and does that result make sense to you?

Please provide the source code as before.